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Deformation pseudo-twinning of the $L2_1$-ordered intermetallic superlattice

Bailey E. Rhodes [a,*], Pulkit Garg [b], Nicolò Maria della Ventura [a], Justin A. Mayer [a], Irene J. Beyerlein [a,b,1], Daniel S. Gianola [a,*]

[a] Materials Department, University of California, Santa Barbara, 93106, CA, USA
[b] Mechanical Engineering Department, University of California, Santa Barbara, 93106, CA, USA

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ABSTRACT

Ordered intermetallic materials are promising candidates for applications in extreme environments due to the strong localized bonding schemes that stabilize the structures against degradation. Here, we investigate the propensity for deformation twinning modes in the $L2_1$-ordered $MnCu_2Al$ intermetallic by inducing a state of severe plastic deformation. *Post-mortem* microstructural analysis reveals the presence of meso-scale deformation twins accompanied by a high degree of nano-scale crystalline heterogeneity. The orientation relationship between the parent and twin structures is consistent with activation of the $\langle 111\rangle\{112\}$ deformation twinning mode. *Ab initio* generalized stacking fault energy curve calculations corresponding to $\langle 111\rangle\{112\}$ twinning shear suggest that the magnitude of the twinning partial is $\frac{1}{4}\langle 111\rangle\{112\}$ and results in the local formation of a metastable orthorhombic phase within the pseudo-twin. This work highlights the discrepancy between the deformation modes of ordered/disordered structures and promotes an increased awareness for the role that long-range chemical order plays in determining deformation modes.

Ordered intermetallic materials are known to possess high strengths and oxidation resistance at elevated temperatures, making them attractive candidates for load-bearing applications in extreme environments [1–3]. Some intermetallic materials are derived from chemically disordered and higher symmetry lattices with relatively large unit cells, thereby leading to large energies required for structure-preserving dislocation slip. In the absence of dislocation slip, deformation twinning modes allow materials to maintain a portion of their ductility and strength, making the discovery of new deformation twinning modes an important scientific objective for materials applied in extreme conditions to prevent brittle modes of failure [4]. At high strain rates, or at low temperatures, the complex dislocation dissociation reactions active in intermetallic materials can lead to an additional deformation mode often referred to as pseudo-twinning [1,4–13].

The parameters of a pseudo-twinning mode are the same as that of a true twinning mode of the same type. They are defined by a shear plane S, twin plane $\mathbf{K}_1$, a conjugate plane $\mathbf{K}_2$, a shear direction η_1, and a conjugate shear direction η_2. Therefore, pseudo-twins are generally expected to have the same crystallographic relationship with the parent as a true twin of the same type, but can unambiguously be identified either by atomic resolution electron microscopy or by the unique pseudo-twin superlattice diffraction peaks. Pseudo-twins will occupy the same volume as, be fully coherent with, and may appear to diffract similarly to true twins. However, true and pseudo-twins are distinct in process and chemistry. True twinning achieves the lattice reorientation through successive shears while preserving the chemical ordering [2,4,14], whereas pseudo-twinning produces the same lattice reorientation through successive shears while creating a new chemical ordering. Thus, for pseudo-twins, the twin shear does not equal that of a true twin and corresponds instead to the lattice vector of the principal metallic lattice upon which the ordered intermetallic is derived [4,5,15]. True twinning modes predicted for some select BCC- and FCC-derived intermetallic phases are presented in Table 1 [1,5]. Most of this table has been replicated from [1] with the exception of the addition of twins in BCC-derived $L2_1$-ordered intermetallics. The geometries of true twins in $L2_1$ intermetallics were defined by Yoo [5]. In that same work, the authors speculated that pseudo-twinning could also occur in $L2_1$-ordered systems. Yet to our knowledge there are no experimental reports of either pseudo- or true twinning activity in these materials.

* Corresponding authors.
E-mail addresses: baileyrhodes@ucsb.edu (B.E. Rhodes), gianola@ucsb.edu (D.S. Gianola).

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Table 1
Geometric parameters of possible true twinning modes in the $L1_2$, $B2$, $B32$, $D0_3$ and $L2_1$ cubic superlattices, as first explored by Christian and Laughlin [1] and Yoo [5]. The mode numbers are based on Bevis and Crocker notation [1,23,24], wherein the first integer of the mode label is a reciprocal of the fraction of atoms sheared to true twin positions in the disordered higher symmetry lattice and the second integer corresponds to the position in the Table presented within [24]. ^{T} denotes transpose modes. S is the shear plane, $\mathbf{K}_1$ is the twin plane, and η_1 is the shear direction. The second undistorted plane (before shear) is $\mathbf{K}_2$ and η_2 is the conjugate shear direction. s^2 is the magnitude of shear necessary to restore chemical order corresponding to the superlattice structure, in terms of the interplanar distance corresponding to $\mathbf{K}_1$. When the magnitude of the twin shear of the twinning mode does not equal the theoretical one, the resulting structure is characterized as a pseudo-twin. Pseudo-twinning is expected to occur when $s^2 > 2$ [2]. Intermetallic phases that have been previously observed to exhibit a given true or pseudo-twinning mode are marked by an asterisk.

Mode	S	$\mathbf{K}_1$	$\mathbf{K}_2$	η_1	η_2	s^2	twin in:
Modes without shuffles							
1.3	(110)	$(1\bar{1}1)$	(001)	$[\bar{1}12]$	$[\bar{1}10]$	2	$L1_2$* [10,25–27]
1.3^T	(110)	$(\bar{1}12)$	$(\bar{1}10)$	$[\bar{1}1\bar{1}]$	[001]	2	$B2$* [6,7,28]
1.9	$(\bar{1}\bar{1}0)$	$(\bar{1}12)$	(001)	$[1\bar{1}1]$	$[\bar{1}10]$	8	$B2$, $B32$, $D0_3$* [11,12,29–32], $L1_2$, $\boldsymbol{L2_1}$
1.9^T	$(\bar{1}\bar{1}0)$	$(1\bar{1}1)$	$(1\bar{1}0)$	$[1\bar{1}\bar{2}]$	[001]	8	$B2$, $B32$, $D0_3$, $L1_2$, $\boldsymbol{L2_1}$
Modes with 50% (non-interchange) shuffles							
2.3^T	(110)	$(1\bar{1}4)$	$(1\bar{1}0)$	$[2\bar{2}1]$	$[00\bar{1}]$	$\frac{1}{2}$	$B2$* [6,28,15]
1.2	(001)	$(1\bar{2}0)$	(100)	[210]	$[0\bar{1}0]$	1	$B2$, $L1_2$
2.5	(001)	$(\bar{1}30)$	(110)	[310]	$[\bar{1}10]$	1	$B2$, $B32$, $D0_3$, $\boldsymbol{L2_1}$
1.3	(110)	$(1\bar{1}1)$	(001)	$[\bar{1}12]$	$[1\bar{1}0]$	2	$B2$, $B32$, $D0_3$, $\boldsymbol{L2_1}$
1.3^T	(110)	$(\bar{1}12)$	$(\bar{1}10)$	$[\bar{1}1\bar{1}]$	[001]	2	$L1_2$

There are, however, many examples of $L2_1$ intermetallics that demonstrate a magnetic shape memory effect where external magnetic fields can cause a martensitic transformation [16,17]. Although $L2_1$-ordered intermetallics are most commonly applied in magnetic applications, they are thought to contribute to strengthening as precipitate strengtheners within high-strength alloys, motivating increased study of their deformation modes [18–22]. Fig. 1 presents the $L2_1$ structure, as well as related BCC- and FCC-derived crystal structures. As shown, $L2_1$ intermetallics are a particular ordering upon a $A2$ (BCC) lattice wherein three elements crystallize with a stoichiometry of XY_2Z, leading to a phase with the symmetry of the $Fm\bar{3}m$ space group [33]. Through careful inspection, it can also be observed that the $A2$, $B2$, $D0_3$, and $L2_1$ are intimately related structures that exhibit different degrees of order on an underlying $A2$ (BCC) lattice. More specifically, the relationship is a hierarchy in symmetry given by $A2 \rightarrow B2 \rightarrow D0_3$ & $L2_1$. In $D0_3$ structures, pseudo-twins, or equivalently the so-called 1.9 true twin mode operating with incomplete twin shear, have been observed [29–32]. Considering that the $D0_3$ and $L2_1$ structures are coequal in the hierarchy of BCC-derived intermetallics (Fig. 1), it is reasonable to expect that the 1.9 twin mode will also be promoted in $L2_1$.

In this letter, we report observation of deformation twins of the pseudo twin mode in $L2_1$-ordered $MnCu_2Al$ when subjected to large plastic strains. We use scanning electron nano-diffraction to characterize twins of the $\langle 111\rangle\{112\}$-type and density functional theory (DFT) calculations to show that the favorable and stable twinning vector corresponds to that of a pseudo-twin.

Bulk polycrystalline $L2_1$-ordered $MnCu_2Al$ samples were prepared through a combination of arc melting and furnace annealing, according to the methods developed by previous investigations [34–37]. High purity starting materials were weighed in a stoichiometric ratio of 1:2:1, totaling a mass of 3.0 g. An additional excess of 10 mg Mn was added to compensate for volatilization during melting. The starting materials were arc melted 4 times in an Ar atmosphere, flipping between melts to ensure homogeneity in the sample. The resulting ingot was wrapped in Ta foil, sealed in an evacuated Si ampule, and annealed at 850° for 1 week, before quenching in water. The ingot was ground by hand into powder, sealed in an evacuated Si ampule, and tempered at 320° for 1 hour. The tempered powder was ground into a fine powder by hand using a mortar and pestle, a form of severe plastic deformation. The powder was sieved through a 140 mesh (109 μm particle size) to isolate particles that had presumably undergone the highest plastic deformation.

An electron transparent lamella was extracted from the deformed powder particles by focused ion beam (FIB) milling with a TFS Helios NanoLab 600, operating at a voltage of 30 kV, and finished with a final ion polishing step at 5 kV, 16 pA. A microstructural analysis was performed using an FEI Talos F200X TEM/STEM, operating at a voltage of 200 kV in standard TEM mode, with a collection angle of 9 mrad. The 4D STEM dataset was collected using a Ceta-M detector, a convergence angle of 0.6 mrad and camera length of 0.125 m. Virtual imaging of the 4D STEM dataset was performed using the LiberTEM Python package [38]. The scientific color map Batlow [39] was used. Further processing of the 4D STEM data and conversion to orientation data were carried out using the py4DSTEM Python package [40] followed by use of EDAX OIM analysis, the orix Python package [41], and Fiji image analysis tools [42]. Simulated diffraction patterns were generated using the JEMS diffraction pattern simulation package [43].

The DFT $\langle 111\rangle\{112\}$ generalized stacking fault energy curves of $L2_1$-ordered $MnCu_2Al$ were calculated within the Vienna ab initio simulation package (VASP) [44] using the Projector augmented wave (PAW) pseudopotentials [45,46]. Exchange-correlations were treated with generalized gradient approximation (GGA) within the Perdew-Burke-Ernzerhof (PBE) formulation [47] and a plane-wave energy cut-off of 500 eV [48]. Spin-polarized calculations were performed and the magnetic moments of Mn, Cu, and Al were initialized with values of $4\mu_B$, $0.5\mu_B$, and $0.5\mu_B$, respectively. The computed lattice parameter of 5.936 Å agrees with the experimental values [37,49,50]. The unit cell was rotated along [111], $[1\bar{1}0]$, and $[11\bar{2}]$ directions and periodically replicated to obtain a supercell with 72 atoms while maintaining stoichiometry. The supercell consists of 18 atomic layers with 4 atoms on each {112} plane and 10 Å of free space, sufficiently large to minimize any interactions between the free surfaces and slip planes. The generalized stacking fault energy curve was obtained by shearing the upper half of the crystal incrementally with linear steps along the [111] direction followed by structural relaxation using a gamma pack k-point mesh of $4 \times 10 \times 1$ while allow-

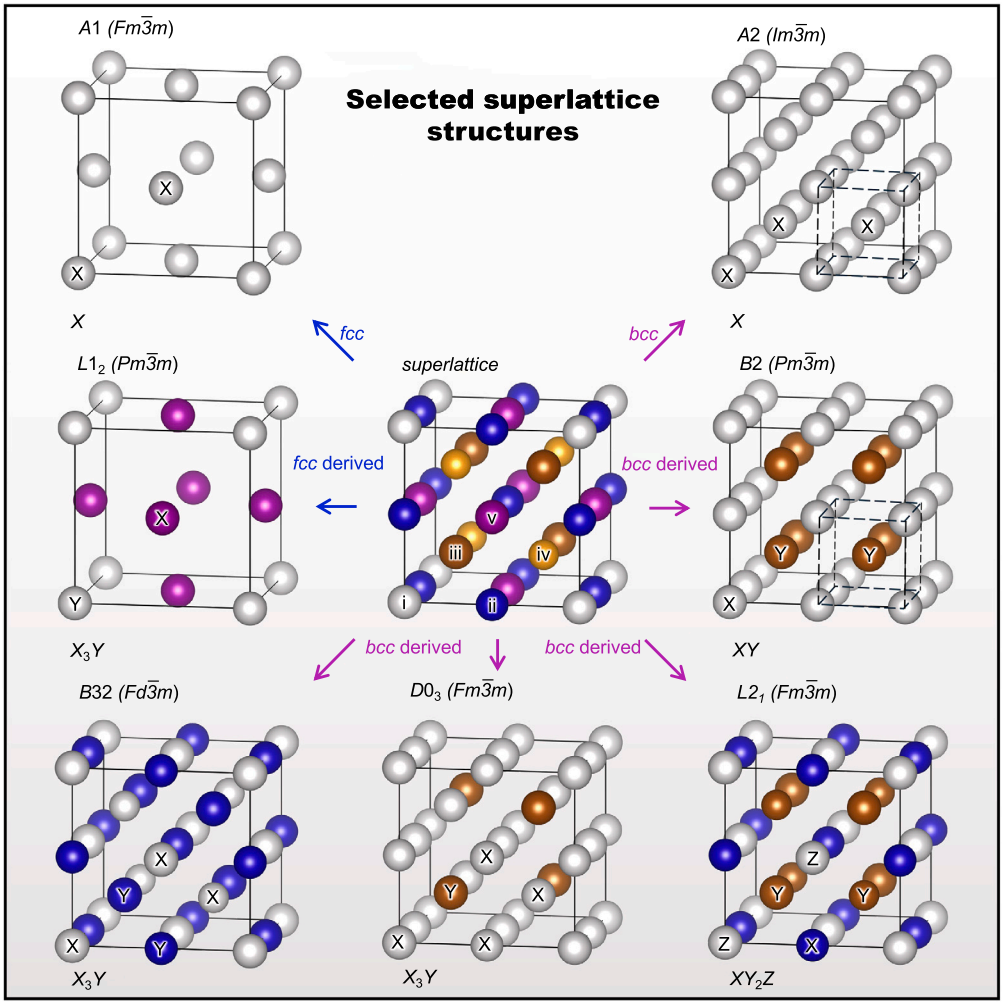

Fig. 1. Schematic showing the relationship between selected FCC- and BCC-based superlattice structures through occupation of unique sites $i - v$. The primitive cells of the $A2$ and $B2$ cells, which are $\frac{1}{8}^{th}$ the volume of the superlattice cell, are indicated with dashed lines.

ing the atoms to relax only along the direction normal to the fault plane [51,52].

We first investigate the impact of severe plastic deformation on the microstructure of $MnCu_2Al$ and investigate the potential formation of true or pseudo-twins within the deformed microstructure. Inspection of Fig. 2a reveals that the microstructure is punctuated by the appearance of banding where intense shear occurs. We focus our analysis on the boxed region of Fig. 2a, where the shear banding is the most prominent. Virtual BF (vBF) and virtual dark field (vDF) STEM generated using virtual annular apertures applied to a 4D STEM dataset collected in this region reveal what appear to be fine-scale needle-like nano domains accompanying the intense shear banding and have maximum widths of 80 nm, yet taper to as slender as 2 - 4 nm in thickness from the view of our imaging (Fig. 2b,c). These features comprise an estimated 11% volume fraction of the field of view. An orientation analysis performed on the dataset in Fig. 2d finds an average out-of-plane orientation near the [001] axis, with spatially varying lattice orientations. Inspection of the spatially averaged diffraction pattern in Fig. 2e reveals azimuthal streaking, which suggests a spread in lattice orientations, likely a result of plastic deformation. The needle-like nano domains observed in Fig. 2a have crystalline orientations that differ substantially from those of the parent grain orientation and are on average oriented with an out-of-plane orientation near [111]. The orientation relationship between the parent and needle-like nano domains can be spatially visualized on a standard stereographic triangle (Fig. 2f). The needle-like nano-domain indicated by a pointed star in Fig. 2d is oriented on average 59° about a $\langle 111 \rangle$ axis, with respect to the matrix crystal (indicated by a star). This orientation relationship is consistent with the expected rotation/axis of the 1.9 mode ($\eta_1 = \langle 111 \rangle$, $\theta = 60°$) [1,53], suggesting that the needle-like nano domains are twins. While the axis/rotation relationship between the crystals is consistent with that of twinning, whether the trace of $\mathbf{K}_1$ coincides with the interface between the crystals needs to be resolved [54].

We next perform boundary trace analysis of the interface between the twinned region indicated in Fig. 2d to identify which of the possible $\mathbf{K}_1$ candidates derived in Table 1 may be activated in the $L2_1$ structure. From the geometric analysis performed by Christian and Laughlin [4] and Yoo [5] in Table 1, it can be assumed that the twin plane must belong to one of the following families of planes $\{111\}$, $\{112\}$, $\{114\}$, $\{120\}$, or $\{130\}$. Within each pole figure in Fig. 3 with both domains marked by stars in Fig. 2d,g overlaid, the plane normals of the given $\{hkl\}$ family corresponding to each respective domain are represented by poles within the stereographic projection. The criteria for an observed boundary trace to be classified as a particular $\mathbf{K}_1$ are: (1) the pole of an $\{hkl\}$ plane of the $\mathbf{K}_1$ type in the parent stereogram coincides with that of the twin stereogram and (2) the observed boundary trace between the twin/parent on the z surface of the foil matches the expected boundary trace associated with the stereographic projection. Coincident poles of the possible $\mathbf{K}_1$ candidates are shown in dashed circles in the respective pole figures and are tabulated in Table 2. The set

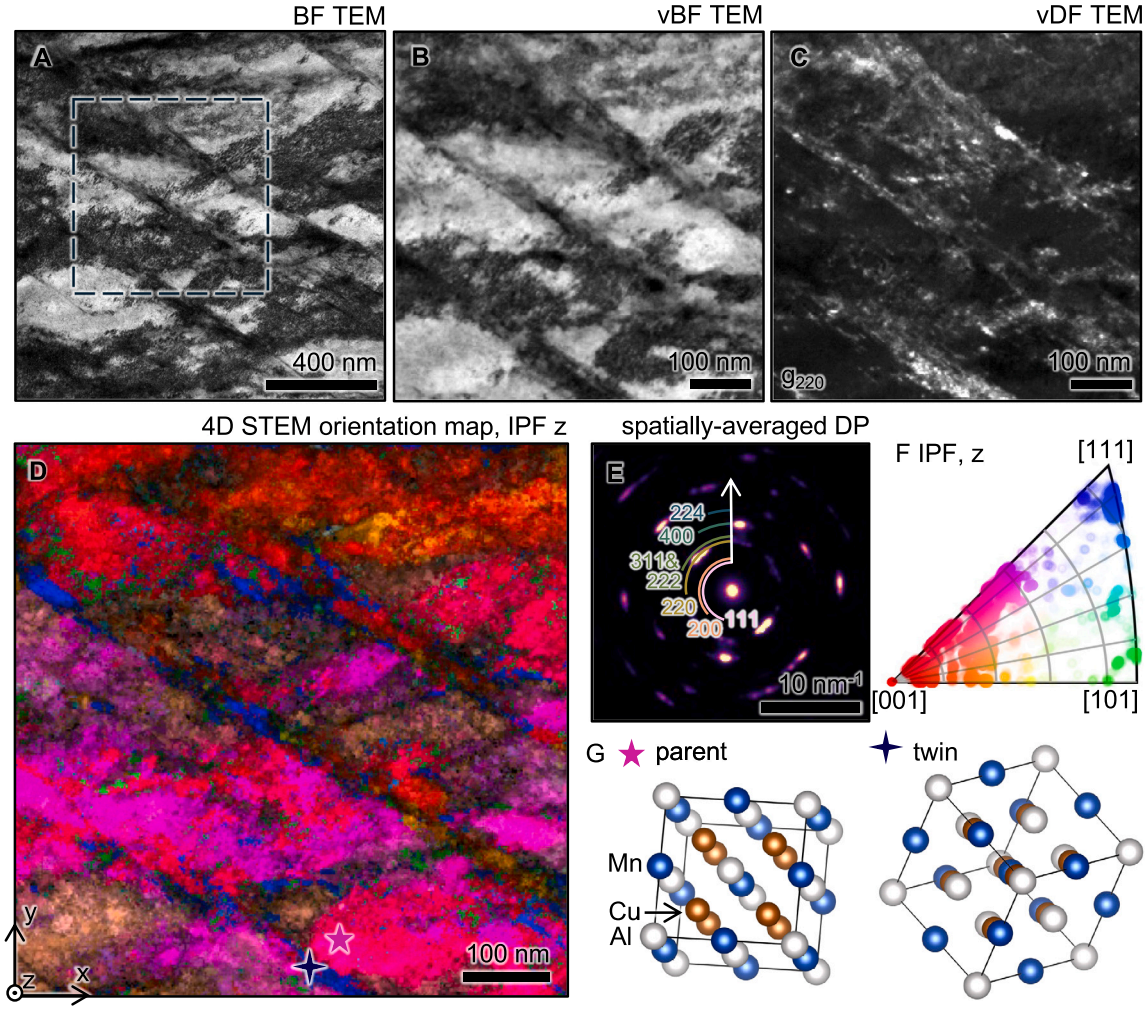

Fig. 2. (**a**) Bright field transmission electron microscope (BF TEM) micrograph. (**b**) Virtual BF TEM micrograph (vBF TEM) and (**c**) virtual dark field TEM (vDF TEM) micrograph generated from a 4D scanning transmission electron microscope (4D STEM) dataset collected from the dashed box in (a). (**d**) 4D STEM orientation map, colored using the inverse pole figure coloring shown in (f) (in the z direction). The intensity of the color of each pixel is scaled using the image quality of the pattern. (**e**) Spatially-averaged diffraction pattern (DP) corresponding to the field of view in (d). From the direct beam out, the indexed diffraction rings correspond to ⟨111⟩, ⟨200⟩, ⟨220⟩, ⟨311⟩ + ⟨222⟩, ⟨400⟩, and ⟨224⟩. (**f**) The $m\bar{3}m$ IPF map in the z direction of (d) shows the distribution in pixel orientations between the [001]-oriented parent toward the [111]-oriented needle-like nanodomains. (**g**) The lattice orientations of the parent and apparent true twinned regions of $MnCu_2Al$ indicated by their corresponding symbols in (d). Subsequent boundary trace analysis is performed between these orientations.

of coincident poles labeled a (⟨112⟩-type pole) are consistent with the observed boundary trace (0.77° deviation, dashed line in Fig. 3a), while the next nearest set of $\mathbf{K}_1$ candidate poles, c (⟨112⟩-type pole) produce a boundary trace 24.5° away, suggesting that $\mathbf{K}_1$ can be confidently identified as belonging to set a.

By definition, the η_1 and η_2 directions of a given twin mode are contained by the $\mathbf{K}_1$ plane and are shared by the two adjoining domains [4]. In stereographic projection, these coincident poles are then expected to lie along the great circle of $\mathbf{K}_1$, which represents the trace of the $\mathbf{K}_1$ plane within the projection sphere. The coincident d (⟨111⟩-type pole) and f (⟨110⟩-type poles overlap the great circle of a (solid line in Fig. 3a – c), and are consistent with the expected η_1 and η_2 shear directions corresponding to the $\mathbf{K}_1 = \langle 112 \rangle$ plane (1.9 twin mode), respectively. Collectively, these crystallographic orientation data are consistent with either true or pseudo-twinning *via* the 1.9 (⟨111⟩{112}) twin mode. While our superlattice diffraction analyses in the Supplementary Information section provide hints of a pseudo-twin structure, knowing the magnitude of the partial dislocations involved in twin formation would help resolve the specific formation pathway.

We employ DFT to identify whether stable partial dislocations exist in the {112} plane of $L2_1$-ordered $MnCu_2Al$ and their magnitude. Fig. 4b shows the generalized stacking faulty energy (GSFE) curve along the twinning direction ⟨111⟩ on the ⟨111⟩{112} plane (as indicated on Fig. 4a). Three local minima exist on this curve, which immediately implies stability of single-plane faults. Notably, the GSFE curve identifies a local minimum associated with a small partial shear occurring at $\frac{1}{4}\langle 111 \rangle$. Achieving this displacement when shearing in the ⟨111⟩ twinning sense requires overcoming a single, relatively small unstable stacking fault of $\frac{3}{20}\langle 111 \rangle$ (0.63 $\frac{J}{m}^2$). The GSFE curve finds that this barrier is smaller than that needed to reach the $\frac{3}{4}\langle 111 \rangle$ in the anti-twinning sense (0.66 $\frac{J}{m}^2$), as well as the $\frac{1}{2}\langle 111 \rangle$ in either the twinning (0.84 $\frac{J}{m}^2$) or anti-twinning sense (0.74 $\frac{J}{m}^2$). The most favorable partial dislocation is therefore the $\frac{1}{4}\langle 111 \rangle\{112\}$ in the twinning sense, which would correspond to the elementary partial for pseudo-twinning.

The stability of $\frac{1}{4}\langle 111 \rangle\{112\}$ suggests that the pseudo-twin in $MnCu_2Al$ can thicken through sequential twinning of adjacent planes, rather than relying, say, on the formation of a multi-plane twin nucleus that must be first stabilized before growth, as is usually the case in BCC alloys [2,8,55]. Fig. 5a constructs the unit cell of $MnCu_2Al$ and Fig. 5b shows its altered configuration after successive, homogeneous shear by the $\frac{1}{4}\langle 111 \rangle\{112\}$ twinning dislocation. The same structure in the [110]

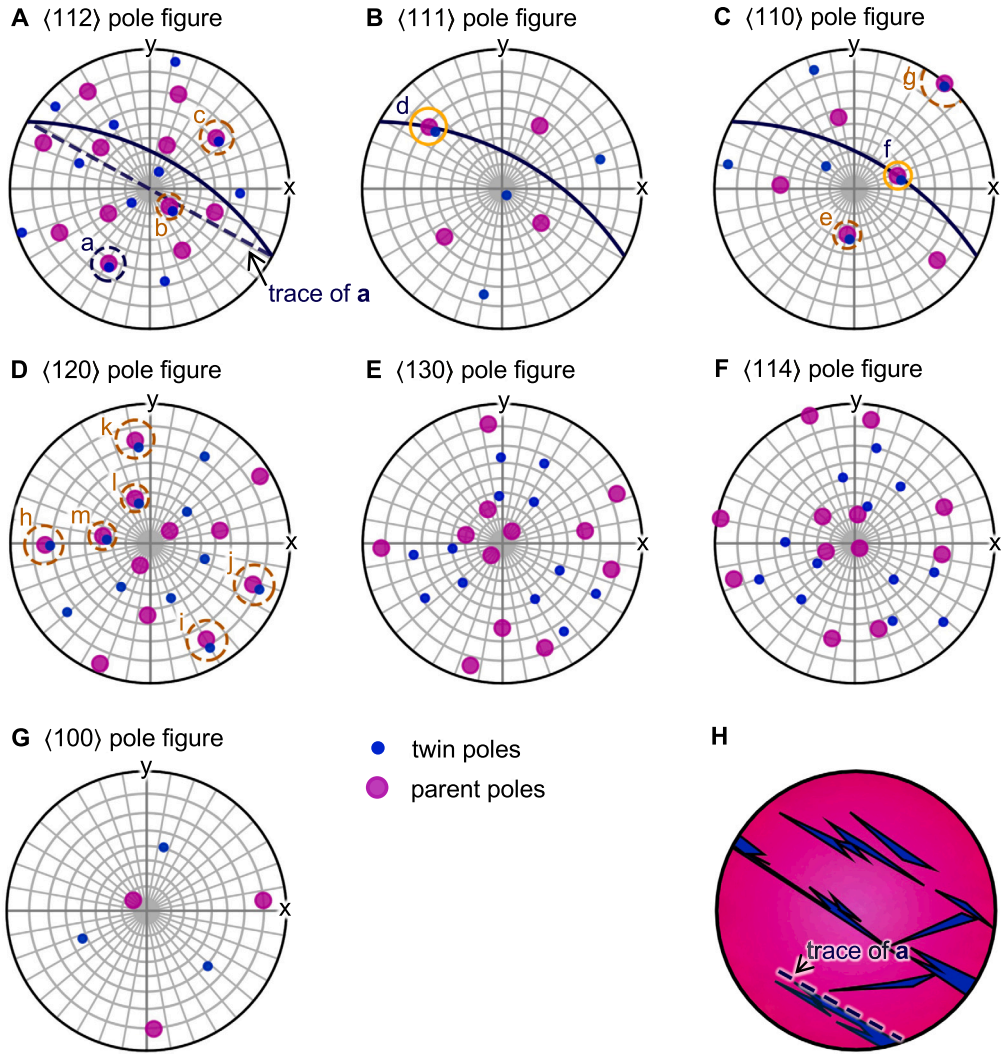

Fig. 3. (**a–g**) Boundary trace analysis for the ⟨111⟩, ⟨112⟩, ⟨114⟩, ⟨120⟩, ⟨130⟩, ⟨110⟩, and ⟨100⟩ poles of planes corresponding the possible twin types observed in the 4D STEM dataset. (**h**) The trace of the {112} plane indicated (*a*) matches the experimentally-observed trace, shown schematically.

Table 2
Analysis of coincident poles in the twin/parent stereograms, indicated by circles in Fig. 3. The observed boundary trace in Fig. 2d is measured as having an expected pole +62° polar angle from the +*x* axis. As a pole 180° polar angle away would be expected to produce an identical trace, we also consider poles with the 242° polar angle.

	pole	misorientation	polar angle
observed	–	–	62° or 242°
a	⟨112⟩	2.31°	241.23°
b	⟨112⟩	4.61°	318.53°
c	⟨112⟩	2.74°	37.54°
d	⟨111⟩	4.46°	139.55°
e	⟨110⟩	3.90°	261.35°
f	⟨110⟩	4.17°	16.99°
g	⟨110⟩	1.21°	49.41°
h	⟨120⟩	2.72°	180.72°
i	⟨120⟩	4.58°	300.38°
j	⟨120⟩	4.08°	338.30°
k	⟨120⟩	4.21°	98.27°
l	⟨120⟩	4.60°	108.63°
m	⟨120⟩	3.90°	170.97°

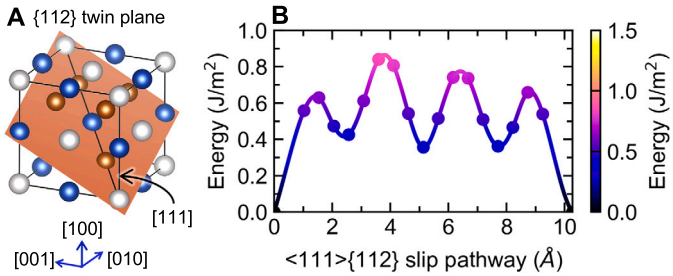

Fig. 4. (**a**) $L2_1$-ordered $MnCu_2Al$, indicating the $\mathbf{K}_1 = \{112\}$ twin plane and $\eta_1 = \langle 111 \rangle$ twin direction. (**b**) Stacking fault energies for $\{112\}$-type planar faults as a function of displacement vector along the $\langle 111 \rangle$ direction. The true twinning mode occurs when the displacement vector is $\langle 111 \rangle$, while a pseudo-twinning mode is predicted when the displacement vector is given by $\frac{n}{4}\langle 111 \rangle$, where $n =$ 1, 2, or 3. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

projection in Fig. 5c enables examination of the pseudo-twin/matrix boundary and the symmetry on either side of it. The pseudo-twinned structure shown belongs to a lower symmetry *Pmmm* space group compared to the $Fm\bar{3}m$ space group of the $L2_1$ parent. We also observe that while in the pseudo-twin, the chemical order of the parent $L2_1$ structure is disrupted, structural site occupation of the underlying BCC lattice is preserved. Analogous metastable structures have been observed or predicted via pseudo-twinning in various $D0_3$ and $B2$ structures [6,9,7,15,28–32]. Our result notably strengthens the connection between these structures and $L1_2$ intermetallics.

In summary, we elucidated the effect of severe plastic deformation on room temperature deformation of the $L2_1$-ordered $MnCu_2Al$ intermetallic using a combination of electron microscopy and computational techniques. Based on our collective findings, the following conclusions can be drawn:

- Electron microscopy identifies $\langle 111 \rangle \{112\}$ deformation twinning in $MnCu_2Al$ when subjected to large plastic strains and suggests the activation of a pseudo-twinning mode.
- DFT calculations provide direct evidence for a stable $\frac{1}{4}\langle 111 \rangle \{112\}$ twinning partial necessary for the pseudo-twinning pathway.
- A new metastable orthorhombic structure is formed by pseudo-twinning $L2_1$-ordered intermetallics by $\frac{1}{4}\langle 111 \rangle \{112\}$ shear on adjacent planes.

These findings not only inform research related to the mechanical operation of $L2_1$-ordered and related intermetallics, but also imply that the planar slip modes operating within the pseudo-twinned structure may be altered in comparison to the non-twinned parent, potentially modifying the composite mechanical performance in materials relying on strengthening from $L2_1$ precipitates. Furthermore, the functional performance of $L2_1$-ordered intermetallics employed in magnetocaloric or magnetic shape memory applications may be influenced by pseudo-twins with unique local magnetic coupling interactions.

CRediT authorship contribution statement

Bailey E. Rhodes: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Pulkit Garg:** Writing – review & editing, Software, Formal analysis, Conceptualization. **Nicolò Maria della Ventura:** Writing – review & editing, Formal analysis, Conceptualization. **Justin A. Mayer:** Writing – review & editing, Software, Conceptualization. **Irene J. Beyerlein:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Daniel S. Gianola:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

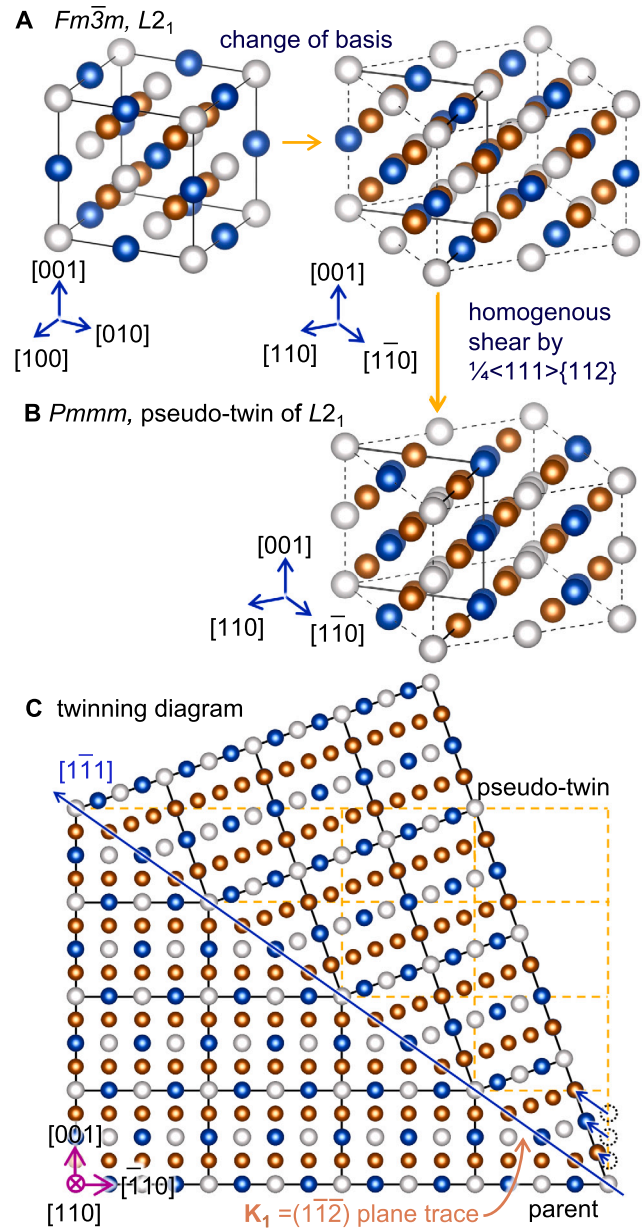

Fig. 5. (**a**) The conventional $L2_1$-ordered $MnCu_2Al$ unit cell in comparison to the $L2_1$ unit cell with a change of basis to show the relationship between the parent and pseudo-twin structures. (**b**) The structure of the predicted pseudo-twin of $L2_1$-ordered $MnCu_2Al$ with space group *Pmmm*. (**c**) Twinning diagram of $L2_1$-ordered $MnCu_2Al$ undergoing shear by $\frac{1}{4}\langle 111 \rangle \{112\}$ partials in the [110] projection. $s = \frac{1}{\sqrt{2}}$.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.scriptamat.2024.116468.

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